



REGULATIONS 2024
FLEXIBLE CHOICE BASED CREDIT SYSTEM
DEPARTMENT OF CIVIL ENGINEERING
B.E. Civil Engineering
CURRICULUM

S.No	Remarks	Category	Course Code R2024	Course Code R2019	Course Title	Prerequisite	Theory Credits	Practical Credits	Total Credits	CGPA / Non - CGPA	Course Type
1	Exempted for LE	HS	SH1106	19EN106	English for Engineers	-	3	1	4	CGPA	TIP (6 Expts)
2	Exempted for LE	HS	SH1107	19EN107	Technical Writing and Presentation	-	0	2	2	CGPA	Practical (10 Expts)
3		HS	SH7801	NA	Human Values and Professional Ethics	-	2	0	2	CGPA	Theory
4	Exempted for LE	HS	SH7101	22HS101	தமிழர் மரபு / Heritage of Tamils	-	1	0	1	CGPA	Theory (Int Only)
5	Exempted for LE	HS	SH7102	22HS102	தமிழரும் தொழில்நுட்பமும் / Tamils and Technology	-	1	0	1	CGPA	Theory (Int Only)
6	Exempted for LE	BS	SH2201	19MA201	Calculus and Matrix Algebra	-	4	0	4	CGPA	Theory
7		BS	SH2211	19MA211	Statistics and Numerical Methods	-	3	1	4	CGPA	TIP (6 Expts)
8	Exempted for LE	BS	SH2224	19MA224	Transforms and Differential Equations	-	4	0	4	CGPA	Theory
9	Exempted for LE	BS	SH3209	19PH209	Physics for Civil Engineering	19PH201 / NA	3	1	4	CGPA	TIP (6 Expts)
10	Exempted for LE	BS	SH4208	19CY208	Basic Engineering Chemistry	-	3	1	4	CGPA	TIP (6 Expts)
11		ES	SH4801	NA	Environmental Sciences and Sustainability	-	2	0	2	CGPA	Theory
12	Exempted for LE	ES	CE0801	NA	Engineering Graphics	-	3	0	3	CGPA	Theory
13	Exempted for LE	ES	CE0802	NA	Engineering Practices Laboratory	-	0	2	2	CGPA	Practical (10 Expts)
14	Exempted for LE	ES	CE0306	19CE306	Engineering Mechanics	-	4	0	4	CGPA	Theory
15		ES	CE0303	19CE303	Fluid Mechanics	-	3	1	4	CGPA	TIP (6 Expts)
16		ES	CE0803	NA	Construction Materials and Technology	-	2	1	3	CGPA	TIP (6 Expts)
17		ES	CE0804	NA	Engineering Geology	-	2	1	3	CGPA	TIP (6 Expts)
18		ES	CS3301	19AI301	Python Programming	-	2	1	3	CGPA	TIP (6 Expts)
19	Any 1	ES	CS3410	19AI410	Introduction to Machine Learning	-	2	1	3	CGPA	TIP (6 Expts)
20	Any 1	ES	CS3801	NA	Artificial Intelligence and Applications	-	1	2	3	CGPA	Theory (Int Only) and Practical (10)
21		PC	CE0401	19CE401	Mechanics of Materials	19CE302 / 19CE306 / CE0306	3	1	4	CGPA	TIP (6 Expts)
22		PC	CE0402	19CE402	Surveying	-	3	1	4	CGPA	TIP (6 Expts)
23		PC	CE0403	19CE403	Soil Mechanics	-	3	1	4	CGPA	TIP (6 Expts)
24		PC	CE0404	19CE404	Applied Hydraulic Engineering	19CE303 / CE0303	2	1	3	CGPA	TIP (6 Expts)

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25		PC	CE0805	NA	Concrete Technology	CE0803	3	1	4	CGPA	TIP (6 Expts)
26		PC	CE0806	NA	Structural Analysis	CE0401	3	0	3	CGPA	Theory
27		PC	CE0407	19CE407	Transportation Engineering	-	2	1	3	CGPA	TIP (6 Expts)
28		PC	CE0418	19CE418	Environmental Engineering	19CE303 / CE0303	3	0	3	CGPA	Theory
29		PC	CE0419	19CE419	Water and Wastewater Analysis Laboratory	-	0	2	2	CGPA	Practical (10 Expts)
30		PC	CE0807	NA	Foundation Engineering	CE0403	3	0	3	CGPA	Theory
31		PC	CE0410	19CE410	Water Resources Engineering	-	3	0	3	CGPA	Theory
32		PC	CE0808	NA	Design of Reinforced Concrete Elements	CE0806	3	0	3	CGPA	Theory
33		PC	CE0412	19CE412	Survey Camp	19CE402 / CE0402	0	2	2	CGPA	Practical (10 Expts)
34		PC	CE0809	NA	Advanced Structural Analysis	CE0806	3	0	3	CGPA	Theory
35		PC	CE0414	19CE414	Design of Steel Structures	19CE406 / CE0806	3	0	3	CGPA	Theory
36		PC	CE0416	19CE416	Estimation, Costing and Valuation	-	3	0	3	CGPA	Theory
37		PC	CE0420	19CE420	Building Information and Modeling	-	0	3	3	CGPA	Practical (10 Expts)
38		PC	CE0810	NA	Computer Aided Plumbing Drawing	-	0	2	2	CGPA	Practical (10 Expts)
39		PC	CE0811	NA	Prestressed Concrete Structures	CE0808	3	0	3	CGPA	Theory
40		PC	CE0812	NA	Robotics and Automation in Construction	-	2	0	2	CGPA	Theory
41		PC	CE0813	NA	Advanced Material Science and 3D Printing	-	2	0	2	CGPA	Theory
42		PE	CE0504	19CE504	Design Studio and Detailing	-	1	2	3	CGPA	Theory (Int Only) and Practical (10 Expts)
43		PE	CE0814	NA	Advanced Mechanics of Materials	-	3	0	3	CGPA	Theory
44		PE	CE0506	19CE506	Industrial Structures	19CE413	3	0	3	CGPA	Theory
45		PE	CE0815	NA	Structural Dynamics and Earthquake Engineering	-	3	0	3	CGPA	Theory
46		PE	CE0816	NA	Bridge Engineering	-	3	0	3	CGPA	Theory
47		PE	CE0817	NA	Rehabilitation and Heritage Restoration	-	3	0	3	CGPA	Theory
48		PE	CE0507	19CE507	Irrigation Engineering and Drawing	-	3	0	3	CGPA	Theory
49		PE	CE0508	19CE508	Groundwater Engineering	19CE410 / CE0410	3	0	3	CGPA	Theory
50		PE	CE0510	19CE510	Integrated Water Resources Management	-	3	0	3	CGPA	Theory
51		PE	CE0511	19CE511	Water Resources Systems	19CE410 / CE0410	3	0	3	CGPA	Theory
52		PE	CE0818	NA	Advanced Hydrology and Flood Risk Management	-	3	0	3	CGPA	Theory
53		PE	CE0819	NA	Hydraulic Modeling and Water Infrastructure Simulation	-	3	0	3	CGPA	Theory

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54		PE	CE0820	NA	Geo-Environmental Engineering	-	3	0	3	CGPA	Theory
55		PE	CE0516	19CE516	Ground Improvement Techniques	-	3	0	3	CGPA	Theory
56		PE	CE0821	NA	Soil Dynamics and Machine Foundations	-	3	0	3	CGPA	Theory
57		PE	CE0822	NA	Rock Mechanics and Tunneling Technology	-	3	0	3	CGPA	Theory
58		PE	CE0823	NA	Geosynthetics and Reinforced Soil Structures	-	3	0	3	CGPA	Theory
59		PE	CE0824	NA	Slope Stability and Landslide Management	-	3	0	3	CGPA	Theory
60		PE	CE0512	19CE512	Geoinformatics Applications	19CE402 / CE0402	3	0	3	CGPA	Theory
61		PE	CE0513	19CE513	Geographical Information System	19CE402 / CE0402	3	0	3	CGPA	Theory
62		PE	CE0515	19CE515	Digital Cadastre	19CE402 / CE0402	3	0	3	CGPA	Theory
63		PE	CE0825	NA	Remote Sensing and Image Processing	-	3	0	3	CGPA	Theory
64		PE	CE0826	NA	Hydrographic Surveying	-	3	0	3	CGPA	Theory
65		PE	CE0827	NA	Photogrammetry	-	3	0	3	CGPA	Theory
66		PE	CE0828	NA	Industrial Wastewater Management	-	3	0	3	CGPA	Theory
67		PE	CE0829	NA	Environmental Quality Monitoring	-	3	0	3	CGPA	Theory
68		PE	CE0520	19CE520	Environmental Impact Assessment	19CE410 / CE0410	3	0	3	CGPA	Theory
69		PE	CE0521	19CE521	Air Pollution Engineering	-	3	0	3	CGPA	Theory
70		PE	CE0522	19CE522	Data Analytics in Environmental Engineering	19CE408	3	0	3	CGPA	Theory
71		PE	CE0523	19CE523	Instrumentation for Pollution Monitoring and Management	-	3	0	3	CGPA	Theory
72		PE	CE0524	19CE524	Airport Engineering	-	3	0	3	CGPA	Theory
73		PE	CE0830	NA	Traffic Engineering and Management	-	3	0	3	CGPA	Theory
74		PE	CE0526	19CE526	Port and Harbour Engineering	-	3	0	3	CGPA	Theory
75		PE	CE0831	NA	Railway Engineering	-	3	0	3	CGPA	Theory
76		PE	CE0832	NA	Pavement Engineering	-	3	0	3	CGPA	Theory
77		PE	CE0833	NA	Transportation Planning Process	-	3	0	3	CGPA	Theory
78		PE	CE0834	NA	Sustainable Construction and Lean Construction	-	3	0	3	CGPA	Theory
79		PE	CE0835	NA	Formwork Engineering	-	3	0	3	CGPA	Theory
80		PE	CE0527	19CE527	Construction Engineering and Occupational Safety	-	3	0	3	CGPA	Theory
81		PE	CE0528	19CE528	Contract Laws and Regulations	-	3	0	3	CGPA	Theory
82		PE	CE0531	19CE531	Construction Project Management	-	3	0	3	CGPA	Theory

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83		PE	CE0532	19CE532	Material Science and Engineering	19CE304	3	0	3	CGPA	Theory
84		PE(Honours 1)	CE0503	19CE503	Prefabricated Structures	-	3	0	3	CGPA	Theory
85		PE(Honours 1)	CE0533	19CE533	Watershed Conservation and Management	-	3	0	3	CGPA	Theory
86		PE(Honours 1)	CE0534	19CE534	Analysis and Design of Deep Foundations	-	3	0	3	CGPA	Theory
87		PE(Honours 1)	CE0535	19CE535	UAV and Drone Technology for Mapping	-	3	0	3	CGPA	Theory
88		PE(Honours 1)	CE0536	19CE536	Solid and Hazardous Waste Management	-	3	0	3	CGPA	Theory
89		PE(Honours 1)	CE0537	19CE537	Intelligent Transport Systems	-	3	0	3	CGPA	Theory
90		PE(Honours 1)	CE0539	19CE539	Advanced Construction Techniques	-	3	0	3	CGPA	Theory
91		PE(Honours 2)	CE0541	19CE541	Precast Technology	-	3	0	3	CGPA	Theory
92		PE(Honours 2)	CE0542	19CE542	Urban Water Infrastructure	-	3	0	3	CGPA	Theory
93		PE(Honours 2)	CE0543	19CE543	Tunneling Engineering	-	3	0	3	CGPA	Theory
94		PE(Honours 2)	CE0544	19CE544	Spatial Data Infrastructure and Management	-	3	0	3	CGPA	Theory
95		PE(Honours 2)	CE0545	19CE545	Water Quality and Management	-	3	0	3	CGPA	Theory
96		PE(Honours 2)	CE0546	19CE546	Urban Infrastructure and Smart Cities	-	3	0	3	CGPA	Theory
97		PE(Honours 2)	CE0547	19CE547	Construction Equipment and Machinery	-	3	0	3	CGPA	Theory
98		OE	XXXXXX	19XXXXX	Open Electives	-	0	0	0	CGPA	Theory
99		OE	XXXXXX	19XXXXX	Online Courses	-	0	0	0	CGPA	Experiential Learning
100		EEC	CE0701	19CE701	Mini Project	-	0	2	2	CGPA	Project (Int Only)
101	Option 1	EEC	CE0702	19CE702	Project Work Phase I	-	0	2	2	CGPA	Project (Int Only)
102	Option 1	EEC	CE0703	19CE703	Project Work Phase II	19CE702 / CE0702	0	6	6	CGPA	Project
103	Option 2	EEC	SH6802	NA	Start Up	-	0	8	8	CGPA	Project
104		EEC	SH6708	19EY708	Career Development Skills	NIL	1	0	1	CGPA	Theory (Int Only)
105		EEC	SH6709	19EY709	Reasoning Ability	NIL	1	0	1	CGPA	Theory (Int Only)
106		EEC	SH6710	19EY710	Quantitative Ability I	NIL	1	0	1	CGPA	Theory (Int Only)
107		EEC	SH6711	19EY711	Quantitative Ability II	SH6710 / 19EY710	1	0	1	CGPA	Theory (Int Only)
108		EEC	SH6712	19EY712	Employment Enrichment Skills	19EY702 / 19EY709 / SH6709	1	0	1	CGPA	Theory (Int Only)
109		EEC	SH6713	19EY713	Placement Readiness Assessment	19EY704 / 19EY711/SH 6711	1	0	1	CGPA	Theory (Int Only)
110		EEC	SH6707	19EY707	Paper Publication	-	0	0	1	NON-CGPA	Experiential Learning (Int Only)

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111		MC	SH8801	NA	Constitutionalism and Artificial Intelligence Policies in India	-	1	0	1	NON-CGPA	Theory (Int Only)
112	Any 1	MC	SH8802	NA	Internship	-	0	0	2	NON-CGPA	Experiential Learning (Int Only)
113	Any 1	MC	SH8803	NA	Entrepreneurship	-	0	0	2	NON-CGPA	Experiential Learning (Int Only)
114	Any 1	MC	SH6803	19MC807	National Service Scheme	-	0	0	0	NON-CGPA	Experiential Learning (Int Only)
115	Any 1	MC	SH8807	19MC808	National Sports Organization	-	0	0	0	NON-CGPA	Experiential Learning (Int Only)
116	Any 1	MC	SH8808	19MC809	Youth Red Cross	-	0	0	0	NON-CGPA	Experiential Learning (Int Only)

BATCH: 2024- 2028 Onwards		
Category	Credits to be Earned (Regular)	Credits to be Earned (Lateral Entry)
Humanities and Science	10	2
Basic Science	22	6
Engineering Science	25	16
Professional Core	62	62
Professional Electives	15	15
Open Electives	12	12
Employability Enhancement Courses	16	16
Mandatory Courses	3	3
TOTAL	165	132